

HARSHUL TANDAN

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EDUCATION

Thunderbird School of Global Management, ASU – MBA in Data Science; GPA: 3.8/4.0 **May 2025**

Relevant Courses: Data Visualisation, Import Export Operations, Business Analytics, Data Mining

Indian Institute of Technology – Master of Technology in Deep Learning; GPA: 3.2/4.0 **Jul 2021**

Relevant Courses: Data Science, Natural Language Processing, Digital Signal Processing

JUIT – Bachelor of Technology in Electronics & Communication Engineering; GPA: 8.2/10 **May 2018**

Relevant Courses: Probability & Statistics, Data Structures, Object-Oriented Programming, Linear Programming Problems

TECHNICAL SKILLS

- **Languages:** Python, SQL, R, Bash, HTML/CSS, Javascript(D3), C, C++, MATLAB
- **Developer/Cloud Tools:** AWS (S3, Glue, Lambda, Redshift, SQS/SNS, API Gateway), Azure (Data Factory, Databricks, DevOps), Snowflake, Git, Docker, Airflow, Tableau, Power BI, Jenkins, CI/CD, Excel, Data Modeling, ETL/ELT Design
- **Frameworks:** Pandas, NumPy, Scikit-Learn, StatsModels, Spark (PySpark), Flask, FastAPI, PuLP/OR-Tools

PROFESSIONAL EXPERIENCE

Protrade Applied Materials Corp. - Supply Chain Consultant; Taipei, Taiwan – Hybrid **Oct 2024 – Dec 2024**

- Collaborated with Finance & Supply Chain teams to elicit and document key requirements for future ESG financial reporting initiatives.
- Performed quantitative analysis (Python, Excel) on sample sustainability data to identify potential cost implications and financial trade-offs for proposed initiatives.
- Developed conceptual data flow diagrams and outlined specifications for foundational Tableau dashboards to visualize proposed ESG metrics.

Thunderbird School of Global Management - Graduate Student Worker; Phoenix, AZ **Mar 2024 – Aug 2024**

- Assisted in LMS content migration by developing Python (Pandas) scripts for data validation, including schema checks and record count reconciliation, to ensure data integrity for academic reporting.
- Provided ad-hoc data analysis and reporting support (Python, Excel) for faculty, generating insights on academic operational metrics.

HDFC Bank - Finance Analytics & Process Improvement Lead; Mumbai, India **Oct 2022 – Jul 2023**

- Conducted portfolio analysis (\$50M) using advanced SQL on large-scale (100M+ rows) financial data in Snowflake; developed Python models (e.g., ARIMA) for capital forecasting.
- Led root cause analysis of cost variances; engineered and automated large-scale data processing workflows using Spark (PySpark) on a Hadoop-based platform (HDFS, Hive for querying) to transform terabytes of financial data, enabling prescriptive analytics and improving operational forecast accuracy by **15%**.
- Designed and deployed self-service Tableau dashboards visualizing financial KPIs and trends, significantly reducing manual reporting effort (**>60%**) and enhancing data accessibility for finance teams.

KYC Hub - Data Pipeline & BI Engineer; London, UK – Remote **Jan 2022 – Jul 2022**

- Engineered automated data ingestion pipelines (Python, AWS Glue, Lambda) handling **>1M records/month** to DynamoDB & S3 data lake; applied dimensional modeling for BI.
- Developed and deployed REST APIs (Python, Flask/FastAPI), containerized with Docker, on AWS Lambda via API Gateway, utilizing SQS/SNS for robust, event-driven asynchronous data processing.
- Implemented CI/CD pipelines (Azure DevOps, Jenkins, Docker) for data and BI components, automating testing (pytest) and deployment to improve release velocity and system reliability.

AI Health Highway - AI/ML Engineer; IISc Bangalore **Aug 2021 – Dec 2021**

- Developed LSTM models (Python, Keras) for cardiac health prediction, achieving 92.23% classification accuracy.
- Championed MLOps adoption: architected CI/CD (e.g., AWS CodePipeline/Jenkins) for AWS model deployment; set up S3 versioning.

EXTRACURRICULARS & PROJECTS

Case Series Club - Director of Engagement & Technology; Phoenix, AZ **Aug 2023 – Present**

- Led ASU's inaugural case competition: managed logistics, marketing, & stakeholder engagement for 20+ teams, 10+ judges.
- Deployed club's first AI chatbot (Python, NLP), automating responses & improving resource access for 500+ members.
- Delivered a featured presentation at the Case AI Conference on innovative AI applications in case study methodology.

Scalable Self-Service AWS Finance Analytics Platform (GitHub) - Proof-of-Concept

- Designed and prototyped an end-to-end BI solution leveraging AWS services: AWS Glue for ETL from S3 data lake, data modeled for Redshift (star schema), and Tableau for visualization. Enabled self-service exploration of simulated SCF data (cost structures, capex, variance) via interactive dashboards with automated data refresh capabilities and alerting (SNS).